Viewpoint

Future of keeping pet reptiles and amphibians: animal welfare and public health perspective

In a review summary on page 450, Pasmans and others discuss the future of keeping reptiles and amphibians as pets. Here, **Clifford Warwick and others** discuss the animal welfare and public health implications of exotic pet business.

IN a review published in this week's *Veterinary Record*, Pasmans and others[1] convey their advocacy for trading and keeping pet reptiles and amphibians. We would argue ethical questions arise around the exotic pet business, not least given the multifaceted global harm involved.[2] In this viewpoint article, we aim to offer a different perspective using the evidence base, focusing on animal welfare and public health.

Animal welfare

Pasmans and others' views on captive reptile health and welfare are unsustainable; the pet industry itself describes 70 per cent mortality at exotics (including reptiles) wholesalers as 'industry standard'.[3] Contrary to Pasmans and others' claims, this abuse is not 'rare'. In an investigation of 15 European wholesalers, 11 were considered prosecutable, because of poor animal welfare standards[4], and other investigations further demonstrate the severity of animal welfare problems in trade environments.[5]

Compared to dogs, which achieve natural longevity in the domestic environment,[6] research has shown up to 75 per cent of reptiles die during their first year in the home.[7] In addition, there are at least 30 behavioural signs of stress regularly observed in captive reptiles, such as hyperactivity, interaction with transparent boundaries, hypoactivity, co-occupant and aggression.[8]

Pasmans and others' faith in certain husbandry information is misplaced. There remains a dearth of independent, objective, scientific evidence-based data concerning the needs of the diversity of species traded and kept. We would argue that the small amount published implies that reptilian and amphibian biological needs are so complex and require such advanced scientific understanding that they cannot be met even in the best zoos (let alone private homes), leading to the basis of 'controlled deprivation'.[9]

Other studies reveal that poor husbandry is common, even for 'well known' species, and that not only is uptake of information poor, but also truth-telling about welfare-relevant issues by keepers is lacking, because an admission of failure to keep alive one's animals is both potentially embarrassing and may involve legal consequences.[10-12] In addition, Moorhouse and others indicate that while public health and legal concerns may dissuade a minority of potential exotic pet purchases, welfare and conservation concerns do not.[13] In brief, the 'five freedoms' (freedom from hunger/thirst, discomfort, pain, fear, and to be able to express normal behaviour) simply cannot be met for captive reptiles.

Public health and safety

Reptiles and amphibians are associated with at least 40 zoonoses.[14] Most

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notorious is reptile-associated human salmonellosis (RAS), although predictably foodborne cases are more common due to proportionality. Pasmans and others fail to mention that 5 to 6 per cent of RAS representation infers approximately 74,000 cases annually in the USA[15] and 6000 cases annually in the UK[7]. In addition, Murphy and Oshin found that 27 per cent of children under five years of age hospitalised with salmonellosis had contracted RAS.[16]

Again, contrary to Pasmans and others' views, decades of extensive guidance relating to hygiene has failed to abate RAS,[17,18] whereas a ban on pet baby turtles (previously responsible for 14 per cent of salmonella infections) in the USA resulted in a 77 per cent reduction in RAS.[19] The prevalence of many other reptile-associated zoonoses has yet to be determined.

Although the number of injuries from reptiles and amphibians may be small, proportionality of ownership is relevant, although the issue remains a growing concern.[20]

Pasmans and others also draw comparisons between dog keeping and reptiles or amphibians. However, dogs are exceptionally affiliative with high-level local (veterinary) expertise available; this is not the case for reptiles or amphibians.

Conclusion

Pasmans and others appear to seek to water down decades of objective science by hundreds of biologists, veterinarians, medics and others. In fact, in very recent years alone, over 500 scientists have produced a veritable library of research and review publications that moderate or contradict Pasmans and others' upbeat claims.

Risk-based species restrictions are indeed valuable, especially before any commercialisation, and not afterwards with subsequent emergent harm, as Pasmans and others would accommodate. The evidence from the highest levels of bioscience demonstrates that mitigating the destructiveness of reptile trading and keeping resides in bans and 'positive lists' (PLs) – species impartially determined suitable to be kept by scientific evidence.

> 'The pet industry itself describes 70 per cent mortality at exotics (including reptiles) wholesalers as "industry standard"'

At least 13 European and 20 North American governments are at various stages of adopting PLs. Evidence-based consumer awareness protocols, such as 'EMODE' (easy, moderate, difficult, extreme),[21] may also prevent irresponsible pet advocacy and acquisition where other measures, including those proposed by Pasmans and others, have failed.

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doi: 10.1136/vr.j4640

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Veterinary Record 2017 181: 454-455
doi: 10.1136/vr.j4640